

Deep Learning-Based Stellar Age Regression Using Astrophysical Observables

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Abstract

Stellar age is a key but non-directly observable quantity in astrophysics, so it must be inferred from correlated stellar measurements. In this project, stellar age estimation for red giant stars is formulated as a non-linear regression problem and solved using a multi-layer perceptron (MLP). The data were taken from the APO-K2 stellar age catalogue through Vizier and include spectroscopic, chemical and asteroseismic measurements [1, 2]. After cleaning physically implausible entries and standardising the inputs, the final model was trained on ten features: effective temperature, spectroscopic surface gravity, metallicity, $[\alpha/\text{M}]$, $[\text{O}/\text{Fe}]$, stellar mass, radius, seismic surface gravity, ν_{max} and $\Delta\nu$. On the held-out test set, the model achieved $\text{MSE} = 0.1709$, $\text{RMSE} = 0.4134$ Gyr, $\text{MAE} = 0.2747$ Gyr and $R^2 = 0.9853$. Additional analyses show stable optimisation, residuals centred close to zero, a median absolute error of about 0.17 Gyr, and physically meaningful feature importance with mass as the dominant predictor. The results demonstrate that a compact MLP can effectively learn the strong non-linear relationships between astrophysical observables and stellar evolutionary age.

1. Introduction

Estimating stellar age is a central challenge in modern astrophysics because age cannot be measured directly for most stars. Instead, it must be inferred from observable quantities linked to stellar structure, composition and evolution. For red giant stars, temperature, gravity, chemical abundances, mass and asteroseismic properties contain rich information about evolutionary state, but their relationship with age is highly non-linear and coupled across multiple physical processes. This makes stellar age estimation a suitable application for deep learning regression.

The aim of this project is to predict stellar age from real observational data using a neural network appropriate for tabular non-linear regression.

1.1. Choice of Network

A Multi-Layer Perceptron (MLP) was selected because the inputs are structured continuous numerical features rather than gridded images or sequences. CNNs are designed for spatial locality, while RNNs are intended for temporal dependencies. Neither assumption applies to this static stellar catalogue. An MLP is therefore the most appropriate of the allowed architectures and is flexible enough to approximate the higher-order interactions between stellar mass, composition and asteroseismic observables.

2. Dataset and Preprocessing

The data come from the APO-K2 stellar age catalogue accessed through Vizier [1, 2]. The raw catalogue contained 1912 stars and 26 columns. Ten physically motivated input variables were selected together with age mode as the target age estimate: effective temperature (T_{eff}), spectroscopic surface gravity ($\log g$), metallicity ($[\text{Fe}/\text{H}]$), α -enhancement ($[\alpha/\text{M}]$), oxygen abundance ($[\text{O}/\text{Fe}]$), stellar mass, radius, seismic gravity, ν_{max} and $\Delta\nu$.

The inclusion of the asteroseismic variables is deliberate. Through seismic scaling relations, ν_{max} and $\Delta\nu$ are linked to surface gravity and mean stellar density, so they provide a direct non-linear constraint on the evolutionary state of a red giant star.

After dropping missing values, removing non-physical ages (≤ 0 or > 20 Gyr), enforcing positive seismic quantities and excluding stars with exceptionally large age uncertainties, the final working dataset contained 936 stars. The data were split into 655 training, 140 validation and 141 test examples.

Figure 1 summarises the cleaned feature distributions and shows that the target age distribution is skewed toward younger stars. This imbalance is relevant for interpretation because it makes regression at large ages more difficult and partially explains why the largest errors occur among the oldest stars.

2.1. Standardisation

The astrophysical features span very different numerical scales, so standardisation was applied to stabilise

Table 1: Representative sample of input data for three stars before (top) and after (bottom) standardisation.

State	T_{eff} (K)	$\log g$	[Fe/H]	$[\alpha/\text{M}]$	[O/Fe]	Mass	Radius	ν_{max}	$\Delta\nu$	Age (Gyr)
<i>Raw data</i>										
Star A	4800.5	2.50	-0.10	0.05	0.02	1.20	10.5	45.2	4.5	4.1
Star B	5100.2	3.10	0.20	-0.02	-0.05	1.50	5.2	120.4	9.8	2.5
Star C	4500.8	1.80	-0.50	0.20	0.15	0.90	25.1	15.6	1.8	9.8
<i>Standardised</i>										
Star A	0.25	-0.15	0.10	-0.20	-0.10	0.50	-0.20	-0.40	-0.35	4.1
Star B	1.10	1.25	1.15	-0.85	-0.95	1.80	-1.15	1.30	1.45	2.5
Star C	-0.95	-1.50	-1.30	1.40	1.55	-1.20	1.85	-1.10	-1.25	9.8

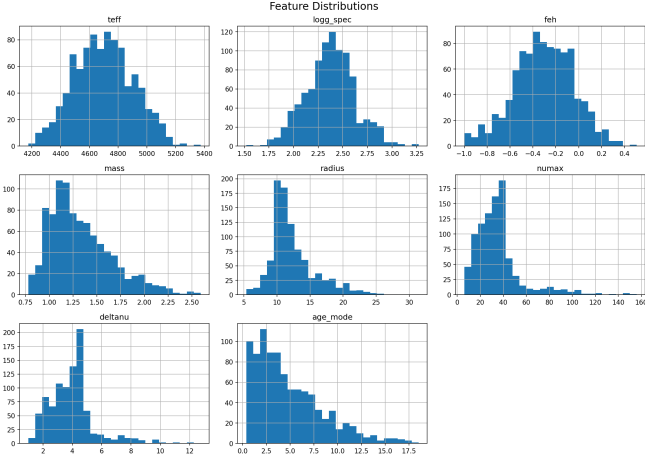


Figure 1: Distributions of the selected input features and target variable after cleaning. The target age distribution is skewed toward younger stars, making large-age regression more difficult.

optimisation and prevent large-magnitude variables from dominating the weight updates. Each feature x was transformed using the training-set mean μ and standard deviation σ :

$$z = \frac{x - \mu}{\sigma}. \quad (1)$$

Table 1 provides a representative before-and-after sample of the preprocessing.

The Pearson correlation matrix in Figure 2 is useful for understanding the physical structure of the data before modelling. Age is most strongly anti-correlated with stellar mass, consistent with the expectation that more massive stars evolve more rapidly. The asteroseismic variables are also strongly correlated with one another because they trace related aspects of stellar structure. Although these trends are informative, the relationships are clearly not purely linear, which motivates a non-linear regression model.

3. Network Design

The final model is a medium-sized MLP comprising an input layer, three hidden layers and a single-node

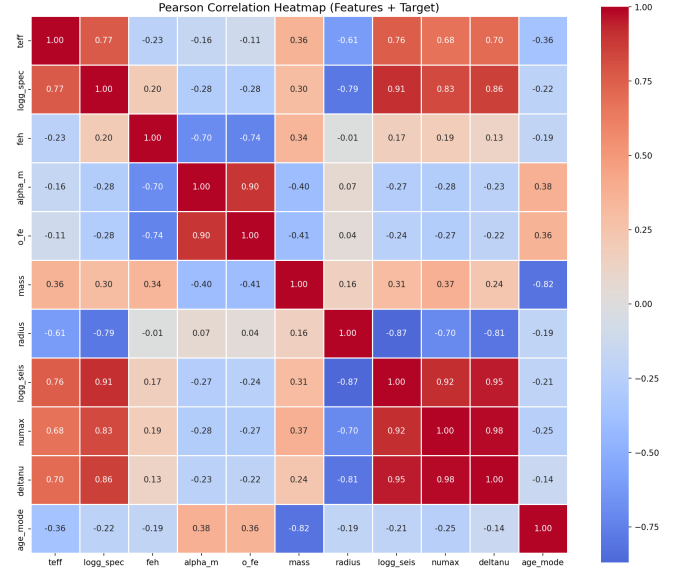


Figure 2: Pearson correlation matrix for the cleaned dataset. Age is most strongly anti-correlated with mass, while the seismic variables are strongly correlated with one another.

output layer.

3.1. Architecture and Activation Functions

The chosen architecture is defined as follows:

$$\begin{aligned} \text{Input} &\rightarrow \text{Dense}(128, \text{ReLU}) \rightarrow \text{Dropout}(0.2) \\ &\rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dropout}(0.2) \\ &\rightarrow \text{Dense}(32, \text{ReLU}) \rightarrow \text{Dense}(1, \text{Linear}) \end{aligned} \quad (2)$$

Rectified Linear Unit (ReLU) activations were used in all hidden layers because they mitigate vanishing-gradient issues and support efficient optimisation in deeper networks. The output node uses a linear activation because the task is continuous regression rather than classification. Dropout layers with rate 0.2 were inserted after the first two hidden layers to reduce co-adaptation between neurons and improve generalisation.

This architecture was selected because it provides sufficient capacity to learn the complex non-linear relationships in the data while remaining compact

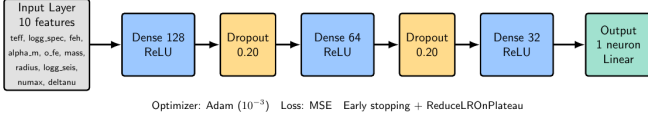


Figure 3: Final MLP architecture used in this work. The network contains three hidden layers with ReLU activations and dropout regularisation, followed by a single linear output node for age regression.

enough to train stably on a dataset of this size.

4. Training Behaviour and Hyperparameters

The network was trained using the Adam optimiser, which computes adaptive learning rates for each parameter and is well suited to heterogeneous tabular data [3]. Mean Squared Error (MSE) was chosen as the loss function because it penalises larger errors more strongly, which is desirable when predicting precise stellar ages.

An initial learning rate of 10^{-3} and a batch size of 32 were used. This learning rate was large enough to allow rapid initial convergence but small enough to avoid unstable oscillatory behaviour, while a batch size of 32 provided a good balance between gradient noise and computational efficiency for a dataset of this size. To reduce overfitting, a dropout rate of 0.2 was applied after the first two hidden layers. In addition, ReduceLRonPlateau was used with patience 10 and reduction factor 0.5 so that the optimiser could take smaller steps once validation performance stopped improving, and EarlyStopping was used with patience 20 and restoration of the best weights to avoid unnecessary late-epoch fitting. These settings were chosen as a practical compromise between fast optimisation, stable convergence and strong generalisation rather than simply maximising training accuracy.

Figure 4 shows that the training and validation loss curves track each other closely throughout optimisation. This indicates stable convergence and suggests that the chosen regularisation strategy successfully controlled overfitting.

5. Test-Set Results and Evaluation

Model performance was evaluated on the unseen test set using MSE, RMSE, MAE and the coefficient of determination (R^2). On the held-out test set, the final MLP achieved $\text{MSE} = 0.1709$, $\text{RMSE} = 0.4134$ Gyr, $\text{MAE} = 0.2747$ Gyr and $R^2 = 0.9853$. These metrics indicate exceptional regression performance, with the model explaining over 98% of the variance in the target ages.

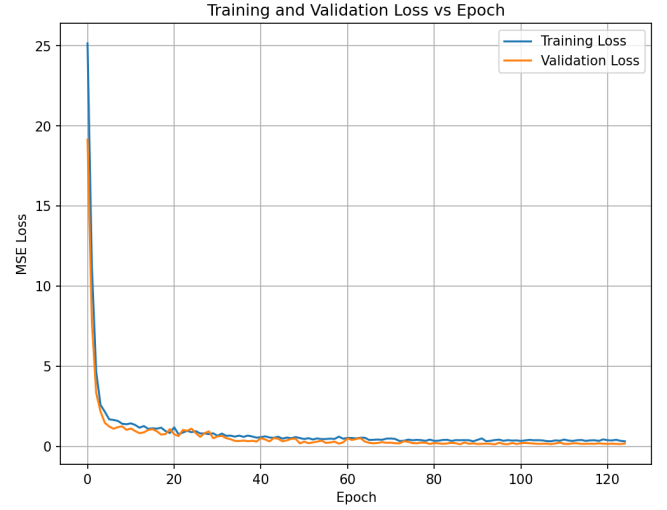


Figure 4: Training and validation MSE versus epoch. Both curves fall rapidly and then flatten, with no strong divergence, indicating stable convergence without severe overfitting.

Figure 5 provides the main prediction-performance plot requested in the brief. The majority of points lie close to the identity line, demonstrating strong generalisation to unseen data. The spread is narrowest for young and intermediate-age stars, while larger deviations emerge for the oldest stars in the sample.

Residual analysis in Figure 6 shows that the errors are centred around zero, confirming that the model does not exhibit a strong global bias toward under-prediction or overprediction. However, the variance increases at larger predicted ages, indicating that the oldest stars are more difficult to estimate accurately.

The cumulative error curve in Figure 7 provides a complementary performance view. It shows that the bulk of predictions are highly accurate and that major deviations are confined to a relatively small subset of difficult cases.

6. Physical Interpretation

A strong scientific model must not only achieve low errors but also learn physically meaningful relationships. To assess this, permutation feature importance and an HR-style comparison between true and predicted ages were used.

The permutation importance analysis in Figure 8 is astrophysically plausible. Stellar mass is the dominant predictor because stellar lifetime depends strongly on mass. Radius and seismic gravity also rank highly because they trace structural expansion along the giant branch.

Mapping both the true and predicted ages back into $(T_{\text{eff}}, \log g)$ space provides a useful physical consistency check. Figure 9 shows that the predicted

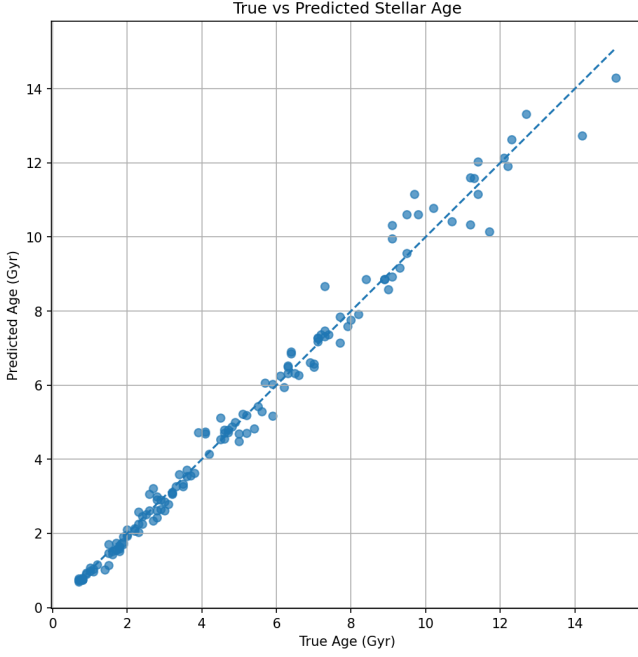


Figure 5: True versus predicted stellar age on the test set. The dashed line marks perfect agreement. Most predictions lie close to the one-to-one relation across the full range.

age field follows the expected macroscopic gradient across the red giant population and broadly mirrors the true HR-style age structure. The main limitation of the model is the heteroscedastic error distribution, with performance degrading slightly for the oldest stars. This is likely driven by data imbalance and by tighter convergence of stellar evolutionary tracks in the upper giant branch, which makes age differentiation inherently harder.

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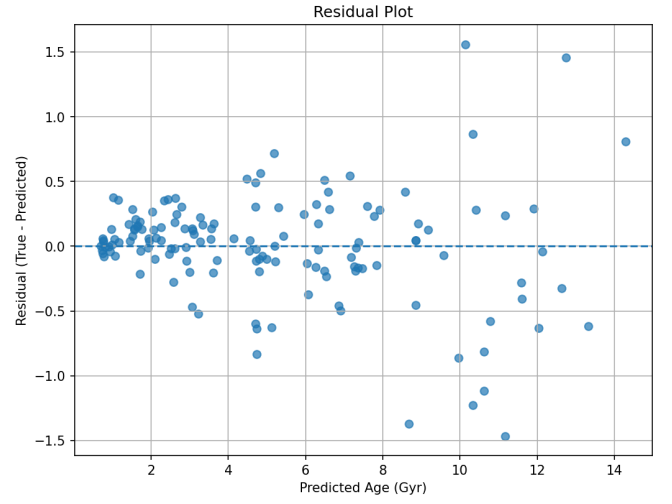


Figure 6: Residuals ($y_{\text{true}} - y_{\text{pred}}$) versus predicted age. Residuals remain centred near zero, but their spread increases for older stars, indicating heteroscedastic uncertainty.

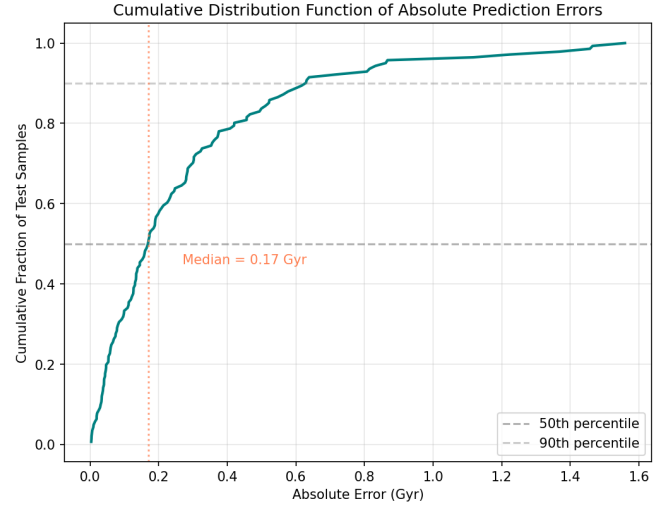


Figure 7: Cumulative distribution of absolute prediction errors. The median absolute error is about 0.17 Gyr, and roughly 90% of test stars lie below about 0.63 Gyr.

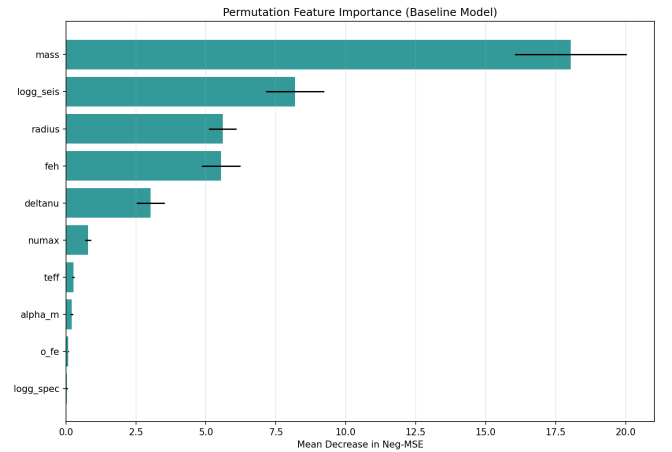


Figure 8: Permutation feature importance for the baseline model. Stellar mass is the dominant predictor, followed by seismic gravity and radius.

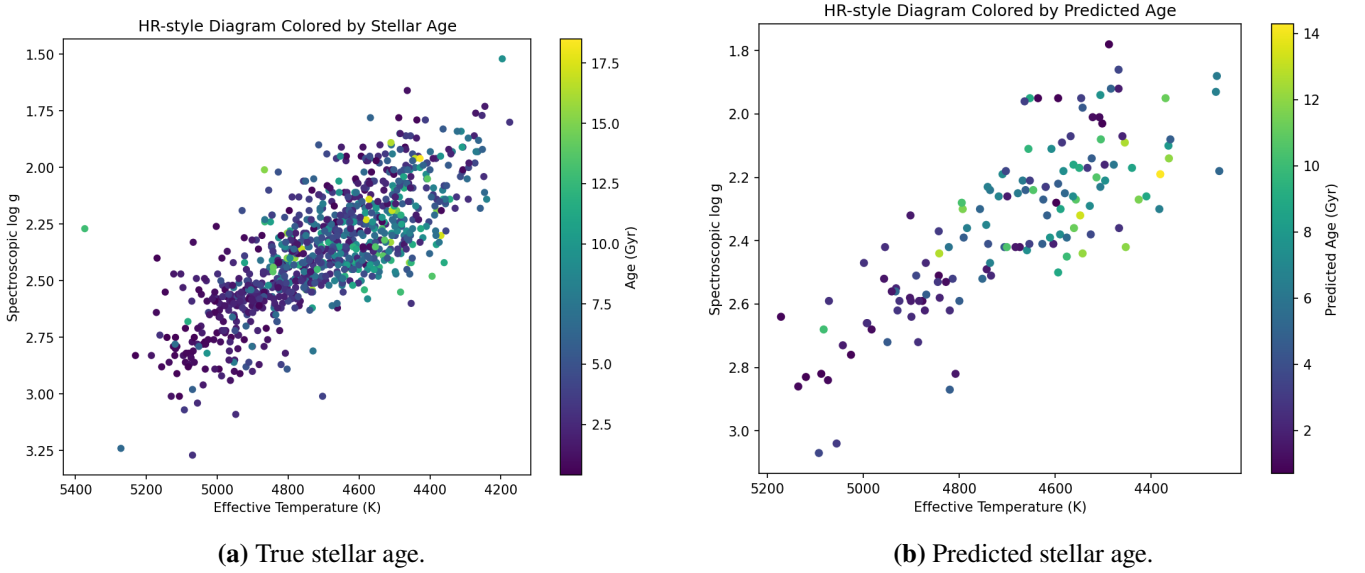


Figure 9: HR-style diagrams in $(T_{\text{eff}}, \log g)$ space coloured by the true ages (left) and predicted ages (right). The predicted map reproduces the broad age gradient seen in the data, indicating that the network has learned a physically meaningful evolutionary ordering rather than merely fitting labels numerically.

7. Conclusion

This project demonstrated that a carefully regularised multi-layer perceptron can perform highly accurate non-linear regression for stellar age estimation using astrophysical observables. The final model achieved $\text{RMSE} = 0.4134$ Gyr and $R^2 = 0.9853$ on unseen test data, trained stably without strong evidence of over-fitting, and produced physically interpretable results through both feature importance and HR-style structure.

Overall, the network performed exceptionally well by both deep learning and astrophysical standards. The results show that deep learning is a powerful tool for stellar chronometry when supplied with high-quality spectroscopic and asteroseismic inputs.

References

- [1] Warfield, J. T., et al. (2024). *The APO-K2 Catalog. II. Accurate Stellar Ages for Red Giant Branch Stars across the Milky Way*. *AJ*, 167, 208.
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- [3] TensorFlow / Keras documentation for Sequential models, Dense layers, Dropout, Adam, EarlyStopping, and ReduceLROnPlateau.